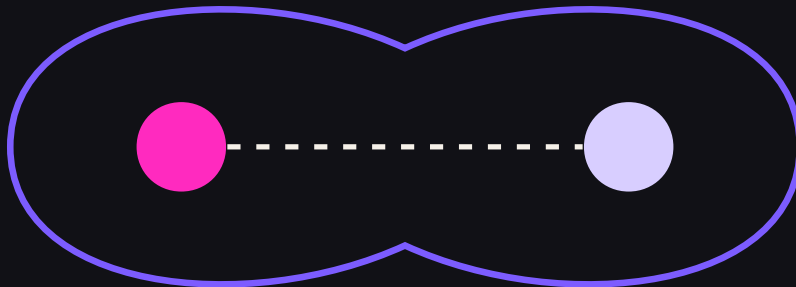


When the Robot Becomes Part of You

How bionics is closing the loop between nervous system and machine

A designed article on robotics, embodiment and the changing boundary between engineered and living systems.



How bionics is closing the loop between nervous system and machine

For a long time, the best prosthesis was the one you could forget you were wearing.

In 2025, two people with above-the-knee amputations tested something different. Their prosthetic knee was anchored directly to bone. Surgically reconstructed muscle pairs in the residual limb generated signals for intended knee movement, and implanted hardware carried them to a powered joint.

They could flex and extend a knee that was no longer biologically there.

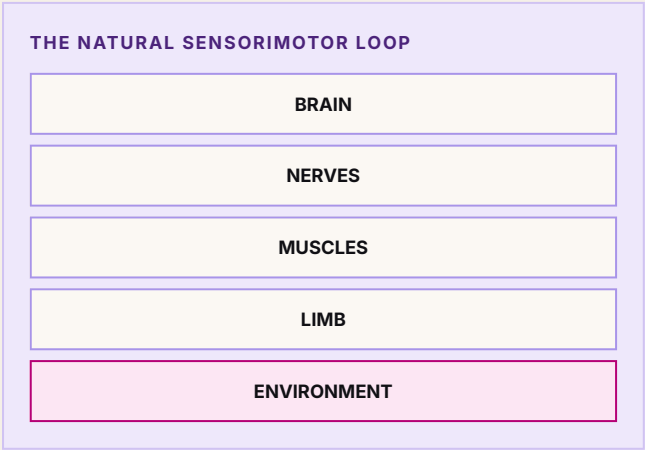
Researchers called it an osseointegrated mechanoneural prosthesis. In the small clinical study it supported walking, obstacle negotiation and sit-to-stand transitions. Participants also reported a stronger sense that the prosthesis belonged to their body.^[1]

That last detail may matter more than the biomechanics. Prosthetics has spent decades improving the machine: lighter materials, better joints, better motors. Bionics is increasingly working on the relationship between machine and nervous system.

A limb plays a different role once intention can flow into it and sensation can flow back. The defining architecture is a loop.

You don't normally operate your leg

Think about walking. You don't calculate joint angles or check that your ankle finished its movement. Your nervous system sends motor signals out while a constant stream of information comes back: muscle stretch, joint position, pressure, touch, balance.



Many prostheses cut that loop. The user sends fairly limited control signals, the device acts, and vision makes up for the sensory information that would normally return through the limb. The result can be surprisingly demanding: sophisticated mechanics, and a person still consciously supervising them.

The frontier is rebuilding the missing connections.

Intention crosses the boundary

One approach starts inside the residual limb. In a conventional amputation, muscles that once worked in opposing pairs lose their natural mechanical relationship, and with it the proprioceptive signals that tell the nervous system where a limb is and how it moves.

The agonist-antagonist myoneural interface, or AMI, surgically reconnects those pairs. In a 2024 Nature Medicine study, seven people with below-knee amputations and AMI surgery used residual muscle signals to continuously control a powered ankle. They were compared with seven people with conventional amputations. The AMI group reached maximum walking speeds 41% higher and adapted the bionic ankle across speeds, slopes, stairs and obstacles.^[2]

■ The machine enters the motor pathway.

Then sensation crosses back

Control gives you half a biological limb. A natural foot also tells the nervous system what is happening at the ground.

In 2024, researchers tested a neuroprosthetic leg that converted pressure under the foot into electrical stimulation of peripheral nerves. They shaped the patterns using computational models of how natural touch receptors respond during contact. Three people with above-knee amputations tried it. When the stimulation more closely copied biological touch, mobility improved and mental effort dropped compared with conventional strategies.^[3]

Sending some signal back is easy. Sending one the nervous system reads intuitively is hard. The closer artificial feedback gets to the language biology already uses, the less translating the user has to do.

Temperature works too. Stimulating particular regions of a residual limb can produce thermal sensations felt in the missing hand. A thin thermoelectric device created cooling sensations in the phantom hand, with mapped regions stable for more than 48 weeks in some participants.^[4] It was then built into a prosthetic hand: a temperature sensor in the artificial finger drove thermal stimulation on the residual limb, and a participant could tell objects apart by temperature felt at the missing hand.^[5]

The sensor sits in an artificial finger. The perception lands in the body. Physical location and perceptual location have come apart, which is exactly the kind of boundary bionics keeps disturbing.

Embodiment happens in layers

Researchers use "embodiment" for how an artificial limb gets incorporated into someone's experience and control of their body. It has several parts:

EMBODIMENT HAPPENS IN LAYERS

AGENCY I experience myself as causing the movement.	OWNERSHIP the limb feels connected to my sense of self.
LOCATION the device occupies the space where I feel my body to be.	FUNCTIONAL INCORPORATION I use it as part of action without treating it as an outside tool.

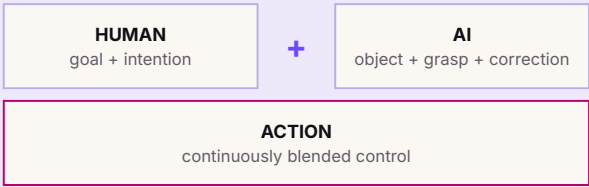
None of these has an on/off switch. The nervous system never officially declares a prosthesis part of the body, and researchers are still working out how to define and measure these dimensions.^[6] Still, mechanical integration moves one boundary, neural control another, sensory feedback a third, and eventually they reinforce each other.

And then AI enters the loop

A biological hand has more than twenty degrees of freedom. Consciously controlling each joint of an artificial one would be exhausting. Our own nervous system handles most of that below the level of attention, and bionic systems increasingly need an artificial version of that division of labour.

A 2025 study fitted a commercial bionic hand with pressure and proximity sensors and shared control between human and machine. The user sent grasp commands through muscle activity. An AI controller used the sensor data to place individual fingers around the object. The two signals were blended continuously. In tests that included four people with transradial amputations, shared control improved grip security and precision and lowered measured cognitive burden.^[7]

SHARED CONTROL



The user still decides to grasp the cup. The hand handles some of the finger-level detail. That should feel familiar, because our biological bodies already work this way: we form an intention, and lower-level systems handle the rest.

Brain-computer interfaces take it further. In 2025, researchers showed a non-invasive EEG interface controlling cursors and robotic arms, then added AI copilots with information about the task. For a participant with paralysis, the AI-assisted system raised target-hit performance 3.9 times in a cursor task. It also let the participant use a robotic arm for a sequential pick-and-place task they could not complete with the interface alone.^[8]

Say I intend to pick up a glass. The neural interface captures my intention imperfectly. AI recognises the object and predicts a grasp. The hand positions its fingers, local sensors adjust pressure, and sensory feedback confirms the grip.

Who picked up the glass? The useful answer is probably that the human-machine system did.

The human changes too

Integration is a process where both sides adapt. In a 2026 pilot randomised trial, 21 people with spinal-cord injury trained with an exoskeleton, either controlled through a brain-computer interface or without it, alongside conventional rehabilitation. The BCI group improved in lower-extremity motor scores, with within-group gains in walking speed and endurance. Some between-group comparisons were not statistically significant, so the results are early, not definitive. EEG also showed changes in cortical activity after training.^[9]

Repeated coupling with a robot may change how the nervous system organises movement. The interface has two adaptive sides.

What happens when nothing is missing?

Most of this work starts from lost or impaired function. The same questions appear in augmentation: can someone learn to control an extra robotic degree of freedom without losing useful control of their own body?

The Third Thumb is a wearable robotic digit mounted opposite the biological thumb. Two motors move it, and pressure sensors under the big toes control it.

In a 2024 study of 596 people aged 3 to 96, 98% manipulated objects with it in their first minute of use. Performance varied with age, but the basic control mapping could be learned across a strikingly broad population.^[10]

Longer training tells you more. In a 2021 study, participants trained for five days. Control, dexterity and coordination with the biological hand improved, and they reported a stronger sense of embodiment. Brain imaging showed changes in how the neural system represented the biological hand, a reminder that adapting to an extra device can change the system it is added to.^[11]

A 2026 study followed participants through seven days. The skill generalised across untrained tasks, body postures, and changes in where the device was worn or how it was controlled. Cognitive demand fell and sense of agency rose. There were trade-offs: toe-based control mildly affected balance, and strong generalisation didn't necessarily make participants choose to use the device when given the option.^[12]

That last point is worth sitting with. Learnability, embodiment, effort, physical cost and voluntary adoption all become part of the engineering problem. Being able to add a capability is not the same as wanting it.

Integration has several boundaries

"Human-machine integration" hides a lot. It helps to separate the layers.

INTEGRATION HAS SEVERAL BOUNDARIES**MECHANICAL**

how physically connected is the technology to the body?

NEURAL

can biological signals directly drive artificial movement?

SENSORY

can the machine write meaningful information back into the nervous system?

COGNITIVE

does using it become ordinary sensorimotor behaviour?

AGENTIC

does AI contribute to the action itself?

The same device can sit at very different points on each layer, and they don't move together. That is why it is so frustrating to look for one precise line between human and machine. There are several, and they move independently.

When AI acts through your body

Shared control raises questions conventional AI rarely makes so personal. If an AI-driven hand adjusts a movement your muscles or neurons started, that correction happens through something you may experience as part of yourself.

Predictability matters. Override matters. Security matters, and so does the privacy of neural and physiological signals. And agency matters, because an effective bionic system still has to feel like you are acting through it.

Good shared autonomy reduces effort without breaking the link between intention and action. That is as much a human-factors problem as an AI problem, and it grows as interfaces improve.

The loop

Prosthetics has a long history of mechanical improvement. The deeper trajectory now shows up in the connections. Bone connects to machine. Muscles generate control signals. Nerves carry artificial sensation. AI fills in motor detail. The nervous system adapts to the whole arrangement.

Describing the technology by where the hardware sits, whether inside the body, attached to it or wrapped around it, tells you surprisingly little about how integrated it is. A better test is whether the loop has closed. Can intention flow out? Can sensation flow back? Can the user act without supervising the device? Can machine intelligence contribute without eroding the sense of agency? Can the nervous system learn to fold all of it into ordinary action?

Physical AI lets machines operate in our world. Bionics raises the possibility that some of them will operate through our bodies. At that point the most important interface may not be the implant, the electrode or the joint. It may be the continuous exchange between nervous system, machine and environment.

If an AI corrected a movement your own nerves started, who would you want to have the final say?

Next: what happens when a machine's body is no longer fixed.

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